Shelter Environmental Checklist
Greening the Shelter

This checklist has been prepared to encourage the use of green and sustainable shelter and settlement approaches in local level planning and implementation. The collected information will act as guidance to further plan for a greener and climate friendly response and reconstruction from local and provincial level by providing status of the existing or planned buildings in reference to environment friendly, greener, and more sustainable practices.

Objectives of the Checklist:

1. Assess the incorporation of environmental criteria focusing on Green Shelter (considering environmental impacts) for buildings and settlement.
2. Recommend practical solutions to make buildings greener and climate friendly with lesser environmental impact during construction and operations.

General Information			
Province		Organization Name	
District		Full Name	
Municipality		Position	
Geographical Region	☐ Mountain ☐ Hilly ☐ Lowland	Gender	☐ Female ☐ Male ☐ Others
Contact Number		Email Address	

Type of Disaster	
Scale of Disaster	**Type of Disaster/Hazard**
☐ Local Level ☐ Province Level ☐ National Level	☐ Earthquake ☐ Landslide ☐ Others ☐ Flood ☐ Fire ☐ Lightning/ Thunderbolt ☐ High Winds

A. Initial Assessment			
A.1. Undertake a broad-level assessment of damage in the sector including environmental factors:			
1.1 Number of households affected in the response area:			
1.2 Information on the socio-cultural nature of housing Typical household size: Typical house size/Area:			
1.3 Damage in Building (Structural component) ☐ Minor (Non-Structural Damage) ☐ Moderate (Some Structural Damage but economically repairable) ☐ Heavy/ Completely Damaged (Structurally Damaged, not economical to repair)			
1.4 Landslide Risk ☐ Minor ☐ Moderate ☐ Heavy	**1.5 Flood Risk** ☐ Minor ☐ Moderate ☐ Heavy	**1.6 Fire Risk** ☐ Minor ☐ Moderate ☐ Heavy	**1.7 Earthquake Risk** ☐ Minor ☐ Moderate ☐ Heavy
1.8 High winds Risk ☐ Minor ☐ Moderate ☐ Heavy	**1.9 Lighting/ Thunderbolt Risk** ☐ Minor ☐ Moderate ☐ Heavy	**1.19 Heat and Cold Waves** ☐ Minor ☐ Moderate ☐ Heavy	

A.2. Review any existing integrated settlement plans or other plans prepared for the area to understand the context of site selection:

2.1 In case of relocation, has the proposed site been identified and mapped by the government for settlement?

Yes No Not Applicable

2.2 Does the planned settlement comply with existing Guidelines/bylaws for housing construction, local technology, and construction materials?

Yes No

2.3 Is the design of shelters considering the possibility of more severe weather conditions?

Yes No

2.4 While allocating sites for shelter, has the needs of the poor and marginalized been prioritized? (Evidence can be participation in planning meetings, etc.)

Yes No

2.5 Information on local climate:

This information will help design the building as per microclimate.
(Temp.: 28℃ in Hills, 40℃ in Lowlands in Summer and 12℃ in Hills, 17℃ in Lowlands in Lowlands, Avg. Precipitation: 600mm)

Temperature:
Precipitation:

B. Community and Affected Population Analysis:

Information on the sociocultural nature of housing

B.1 Predominant housing typology (Multiple Choice)

☐ SMM (Stone Masonry Mud Mortar) ☐ SMC (Stone Masonry Cement Mortar)
☐ BMM (Brick Masonry Mud Mortar) ☐ BMC (Brick Masonry Cement Mortar)
☐ RCC (Reinforced Cement Concrete) ☐ Others

Please Specify Others:

B.2 Number of affected vulnerable households: Click or tap here to enter text.

B.3 Choose any of the following showing household diversity:

Disabled Single Mother Heavy Child
Senior Citizenship Marginalized None Other

Please Specify Others:

B.4 Presence of community-based organizations:

Yes No

B.5 Community engagement

Local youth club User committee Women Community Group
Working group Others.

Please Specify Others:

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B.6 Availability of skilled human resources for shelter design and construction

☐ Engineer ☐ Sub- Engineer ☐ Overseer ☐ Mason ☐ Others

Please Specify Others:

B.7 Available natural resources

☐ Stone ☐ Wood ☐ Sand ☐ Aggregate ☐ Bamboo ☐ Other

Please Specify Others:

B.8 Pre-identified site for debris disposal and solid waste management:

Yes No

B.9 Has guideline for Debris management been established for efficient resource management incorporating reuse, recycle, or repurpose:

Yes No

C. Planning and Design:

C.1 Type of Proposed Site

☐ Emergency Shelter ☐ Existing Permanent Shelter ☐ proposed site for Relocation ☐ Other

Please Specify Others:

C.2 Do the proposed sites affect the natural water resources and supplies?

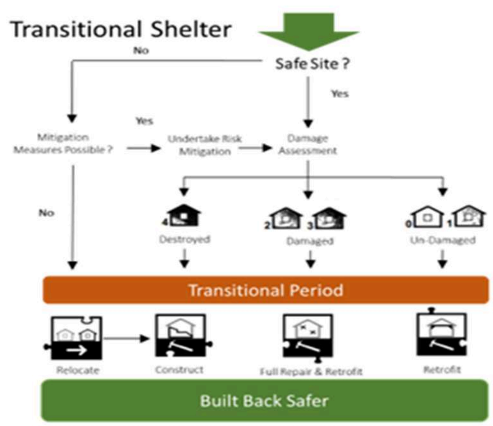
Yes No

If yes, what can be the mitigation measures:

C.3 Is the proposed construction site prone to landslides, soil erosion, and flooding?

Yes No

If yes, what can be the mitigation measures (Incorporate safety features to lessen the impact of hazards, such as raised floors if flooding is a risk, adherence to building safety codes, and use of light building materials in case of earthquakes.)



C.4 Do the proposed sites affect the natural water flow(watershed) such as stream lakes or rivers?

Yes No

C.5 Does the proposed shelter design align with existing building housing typology i.e., locally, and culturally appropriate designs?

Yes No

C.6 Does the shelter planning incorporate measures such as planting trees and other vegetation to provide shade, and installing green roofs to help control heat gain and minimize cooling demands?

Yes No

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D. Construction Materials:

D.1 Which of the construction materials has been used majorly?

Natural Construction Materials:	Processed/Man-Made Construction Materials:
☐ Stone ☐ Timber ☐ Bamboo ☐ Slate ☐ CSEB Tiles ☐ Mud ☐ Clay Brick ☐ Thatched Straw ☐ Others Please Specify Others:	☐ Brunt Brick ☐ Concrete ☐ Glass ☐ Paints ☐ Composite Pa ☐ Ceramic Tiles ☐ Aluminum composite panel ☐ Wall Papers ☐ Autoclaved Aerated concrete ☐ Rebar ☐ Corrugated Galvanized Iron sheets. ☐ PVC Panels ☐ Others Please Specify Others:

D.2 Which of the construction materials can be reused and recycled for reconstruction?

D.1 In case of relocation, has the proposed site been identified and mapped by the government for settlement ☐ Wood ☐ Stone ☐ Clay Brick ☐ Mud Brick ☐ Timber ☐ Brunt Bricks ☐ CGI Sheets ☐ Crushed Concrete Aggregates ☐ Other

Please Specify Others:

D.2 Has the use of harmful materials such as toxic paints and plastics been planned for construction?

⬭ Yes ⬭ No

E. Energy efficiency and building comfort:

E.1 Is the site microclimate considered for the building's location and positioning? (Locate buildings in favorable sites where possible e.g., sheltered areas in cold climates, and exposed to prevailing wind direction in hot climates)

⬭ Yes ⬭ No

E.2 Does the building design provide solar gain during winter?

⬭ Yes ⬭ No

E.3 Does the Building Design provide shading in the summer?

⬭ Yes ⬭ No

E.4 Are there sufficient wall openings (doors and windows) to provide cross ventilation in the building/design?

⬭ Yes ⬭ No

E.5 Does the roofing material have a design that helps to keep buildings cool during summer?

⬭ Yes ⬭ No

E.6 Is there a provision for planting or safeguarding deciduous trees in the planning process? (To promote shading and prevention of soil erosion)

⬭ Yes ⬭ No

E.7 Are low-energy lights such as LEDs?

⬭ Yes ⬭ No

E.8 What are the existing renewable energy sources being used?

☐ Solar Energy ☐ Hydro ☐ Biogas ☐ Other

Please Specify Others:

E.9 What are the planned renewable energy sources for the building?

☐ Solar Energy ☐ Hydro ☐ Biogas ☐ Other

Please Specify Others:

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E.10 Is the inclusion of renewable energy considered during the planning phase or not?

◯ Yes ◯

E.11 What type of cooking fuel and stoves are used mostly? (For e.g., Induction, Rice cooker, etc.)

☐ Firewood ☐ Electricity ☐ LPG ☐ Solar ☐ Other

Please Specify Others:

E.12 Are there any plans to reduce or change the current use of Firewood into Biogas or LPG?

◯ Yes ◯ No

E.13 Does the building design and planning consider the potential risks of extreme weather events and include measures to reduce the number of items or elements exposed to uprooting on the roofs and sides of the building?

◯ Yes ◯ No

F. Water efficiency:

F.1 Is there an existing provision for a rainwater harvesting system or has such provision been planned in the building?

◯ Yes ◯ No ◯ Not Applicable

F.2 Has the use of renewable water heating systems been embraced or not?

◯ Yes ◯ No ◯ Not Applicable

F.3 Is there any system that has been implemented or is being planned to decrease water usage such as greywater harvesting?

◯ Yes ◯ No ◯ Not Applicable

F.4 Are there any considerations or planning for the multiple uses of water (water use plan) at the HH and community level?

◯ Yes ◯ No ◯ Not Applicable

G. Sourcing, use, and disposal of construction materials:

G.1 Is there a possibility to use/ utilize local skills and materials for construction?

◯ Yes ◯ No ◯ Not Applicable

G.2 Is there a possibility to incorporate principles of green logistics and procurement in the construction? (Procure from near marketplaces in bulk to minimize carbon emissions)

◯ Yes ◯ No ◯ Not Applicable

G.3 Have construction materials (sand, Gravel, Boulders, stone, cement, rebars) been procured from legally recognized quarries and entities with minimum environmental impact?

◯ Yes ◯ No ◯ Not Applicable

G.3 Has timber for construction been harvested legally and responsibly from national and/or community forests or private land?

◯ Yes ◯ No ◯ Not Applicable

G.1 Has the softwood timber been treated against insect attack for longer life using safe products?

◯ Yes ◯ No ◯ Not Applicable

G.1 Has efficient building design typology considered waste and cost?

◯ Yes ◯ No ◯ Not Applicable

G.1 Are there any provisions for efficient and safe Debris Management?

◯ Yes ◯ No ◯ Not Applicable

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H. Good Construction Practices:

H.1 Has proper site protection work been designed for the construction sites?

Yes No Not Applicable

H.2 Have proper health and safety practices guidelines been established for construction sites?

Yes No Not Applicable

H.3 Has training been provided to masons and carpenters on the best construction practices for health, safety, and environment?

Yes No Not Applicable

H.4 Are there any proper storage facilities segregated as hazardous, flammable, toxic, etc. materials in the construction site?

Yes No Not Applicable

H.4 Are there any preventative measures incorporated into the design of buildings to protect against moisture damage and termite infestation, such as overhangs, porches, drainage systems, and barriers specifically designed for termites?

Yes No Not Applicable

I. Monitoring and Evaluation:

I.1 Does the construction and building design comply with the existing National Building Code and bylaws?

Yes No Not Applicable

I.2 Has legal and sustainable sourcing of materials been carried out?

Yes No Not Applicable

I.3 Have health and Safety guidelines been implemented through Health and Safety, Environmental checks monthly?

Yes No Not Applicable

I.4 Have community audits and monitoring for construction quality and environmental impacts been carried out in regular intervals?

Yes No Not Applicable

I.5 Has Community feedback been collected and incorporated into design, construction, and monitoring?

Yes No Not Applicable

I.6 Have the measures incorporated in the above sections been evaluated for the effectiveness of environmental measures and their impacts?

Yes No Not Applicable

I.7 Type of Feedback received from the community regarding the overall planning and construction?

Positive Negative Constructive Other

Please Specify Others:

I.8 Has a mechanism been developed to document the lessons learned and incorporate the learnings into the next response?

Yes No Not Applicable

I.9 Has the project considered the steps to comprise gender and diversity in all stages such as planning, implementation, and monitoring?

Yes No Not Applicable